

Sales Analysis Dashboard Report

Introduction:

This project focuses on analysing company sales data using Power BI. The project analyses overall sales performance, customer behaviour, discount & profitability, regions & market, operational efficiency and product performance. The dataset includes orders from 2016 to 2019 with details such as sales, profit, discount, customer details, category information and product information. The main objective of this project is to clean the data, perform transformations using Power Query, and build interactive dashboards that help in decision-making through meaningful visualizations.

Methodology:

The project was developed in Power BI following these major steps:

1. Data Loading: Imported the sales dataset into Power BI.
2. Data Cleaning & Transformation:
 - Removed duplicates based on Order ID.
 - Created new columns such as Year (from Order Date).
 - Filtered data for the year 2019 and for orders with discounts.
3. Data Modeling: Established relationships between tables and created DAX measures such as Total Sales, Total Quantity, Average Sales, Average Profit etc.

4. Visualization: Built multiple charts like bar, line, pie, card visuals to represent insights.
5. Dashboard Creation: Combined visuals into one interactive dashboard for analysis.

Requirement Analysis:

1. Software Requirements:

- Power BI Desktop
- Microsoft Excel for data

2. Data Requirements:

- Fields used are Order ID, Customer Name, Profit, Discount, Region, Segment, Category, Sub-Category, Product name, Quantity and Sales.

3. Analytical Requirements:

- Analyse total sales, profit, and quantity trends.
- Compare before and after removing duplicate orders.
- Identify top-performing products and customers.
- Track sales growth over years and months.
- Understand the impact of discounts on sales.

Data Preparation and Transformation:

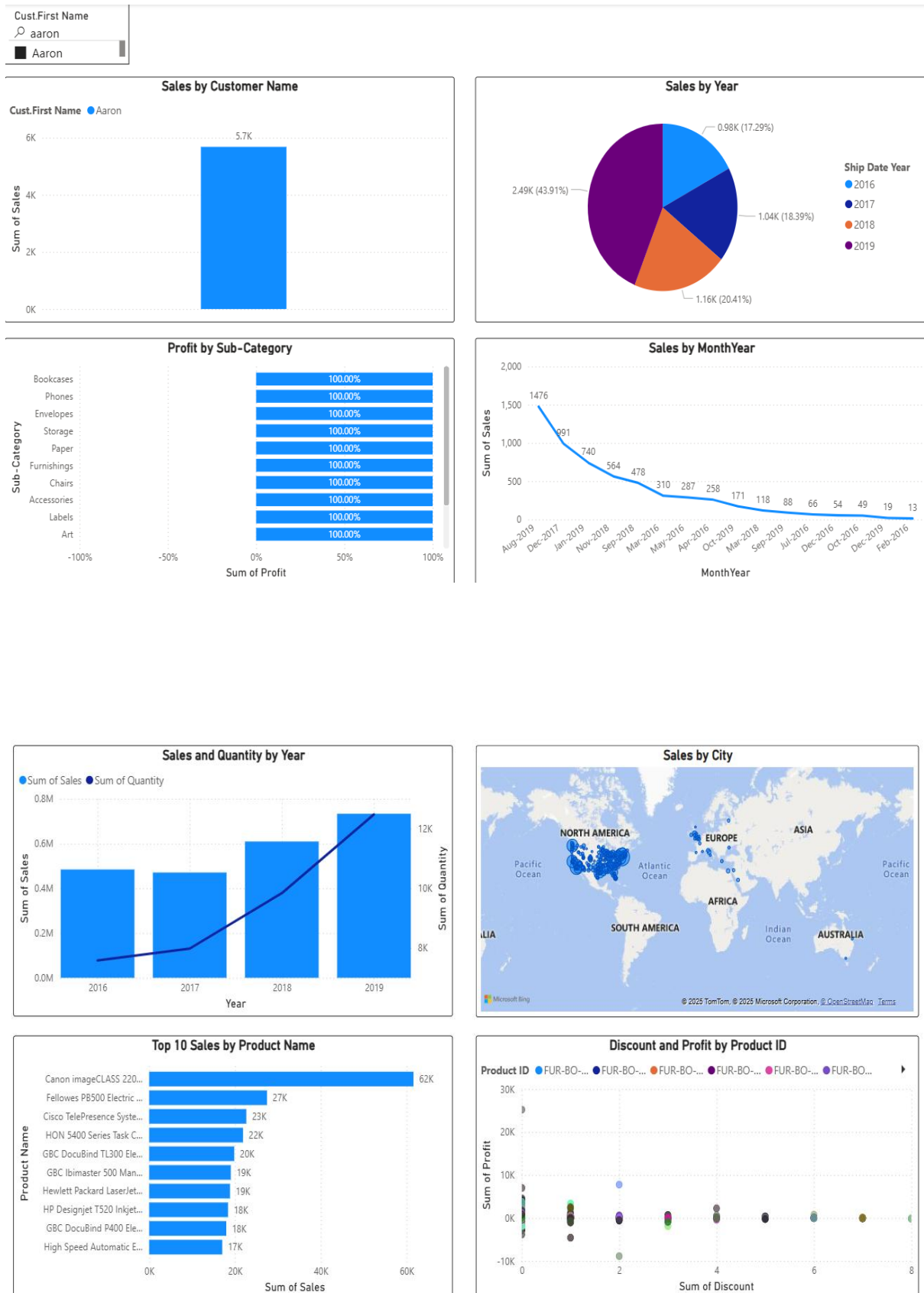
In this project, several data transformation steps were performed in Power Query and through DAX calculations to prepare the dataset for analysis.

Each transformation or parameter was designed to improve data accuracy and enhance insights through Power BI visualizations.

The following operations were done as a part of data transformation:

- ✓ **Year Extraction:** The year was extracted from order date using power query. It was added into a new column called Year.
- ✓ **Filtering Year 2019:** The data for the year 2019 was filtered using power query to obtain insights about that particular year alone. This was done by referencing the source data.
- ✓ **Removed duplicated based on Order ID:** The duplicates were removed based on Order ID for better accuracy. A comparison of total sales before and after duplication was visualised. The source data was referenced as a new table and the duplication was removed in that particular table.
- ✓ **Discount based filtering:** Discounts greater than zero was filtered out, to see the impact of discounts on sales.
- ✓ **Grouping and Aggregation:** Data was grouped by year and the average of sales.
- ✓ **DAX calculations:** Several measures were calculated using DAX such as Total sales, Total Profit, Total Quantity, Average Sales, Average Profit, Average Quantity, Sales after removing duplicates.
- ✓ **Visualization:** Different kinds of charts were prepared as per the need of insights to be drawn from it. Charts were prepared by giving consistent colour schemes and data labels. Titles were also added for better understanding.

Visualizations:



2,297.20K

Total Sales

0.09

Average discount

0.01K%

Profit Margin

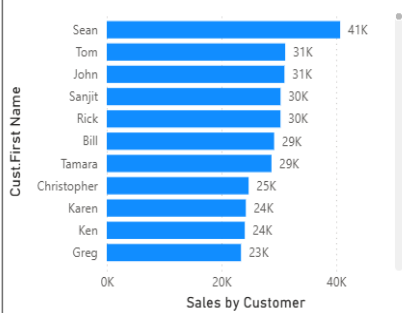
286.40K

Total Profit

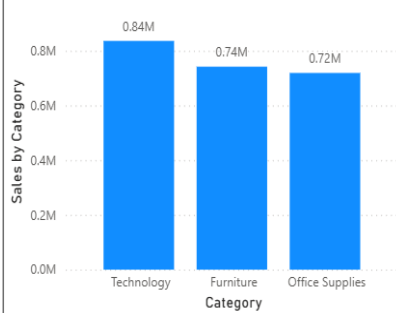
38K

Total Quantity

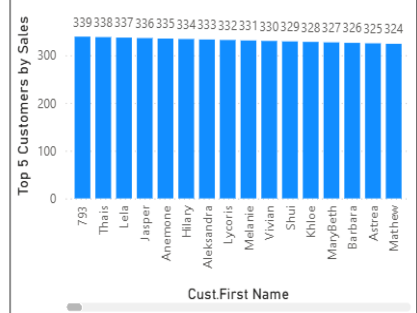
Sales by Customer by Cust.First Name



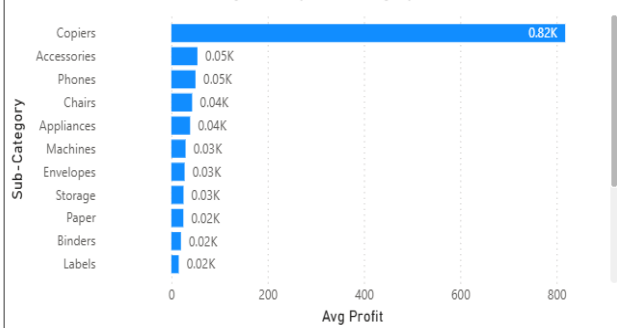
Sales by Category by Category



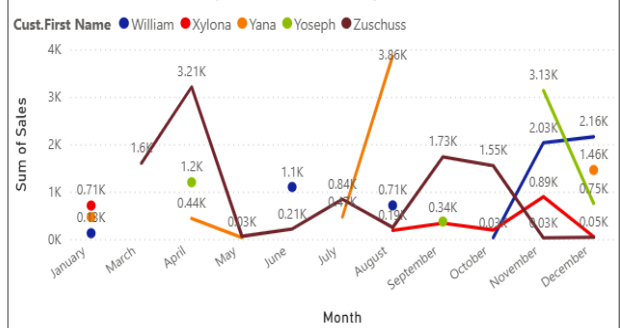
Top 5 Customers by Sales by Cust.First Name



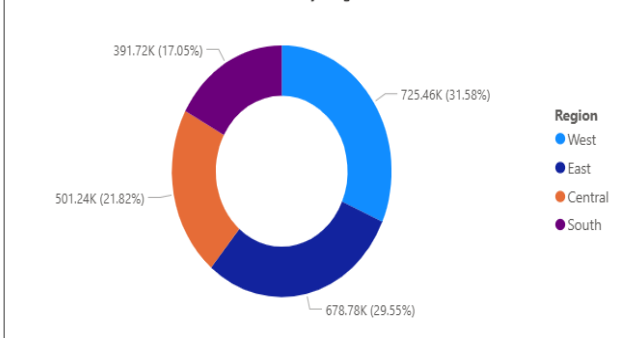
Avg Profit by Sub-Category



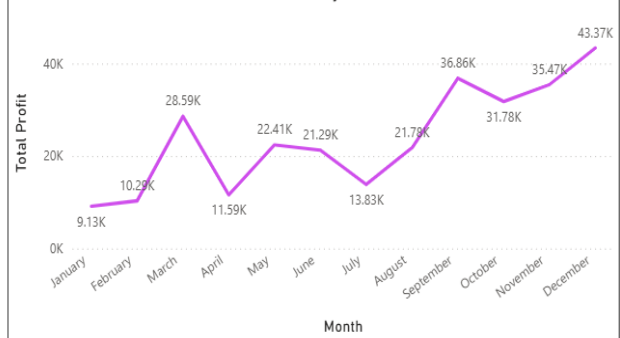
Top 5 Customer Sales by Month

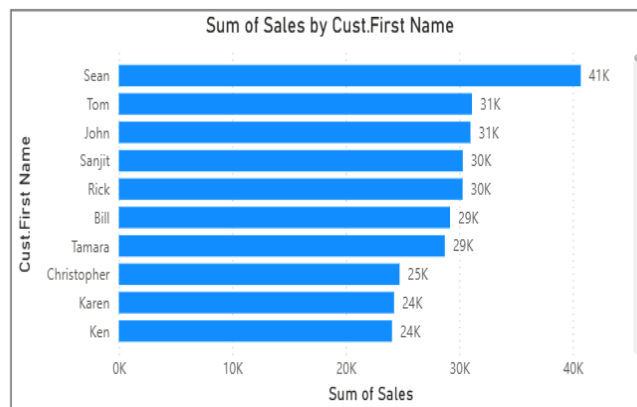
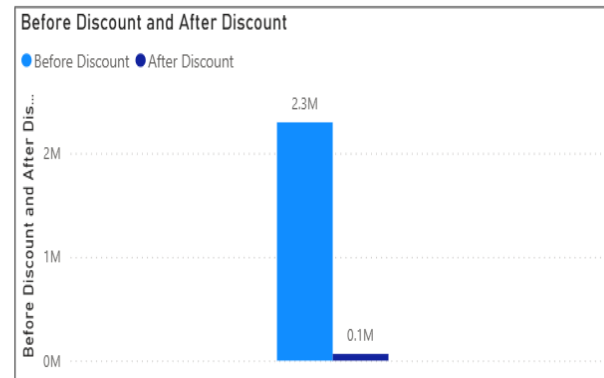
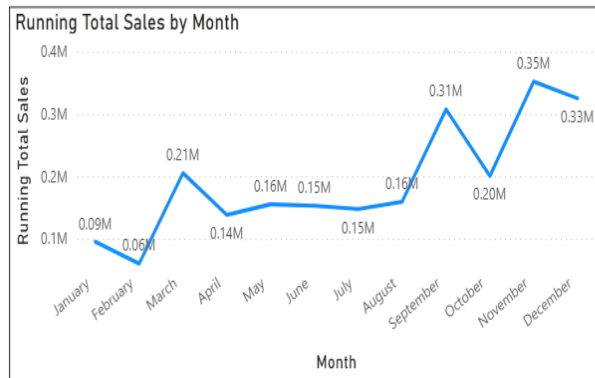
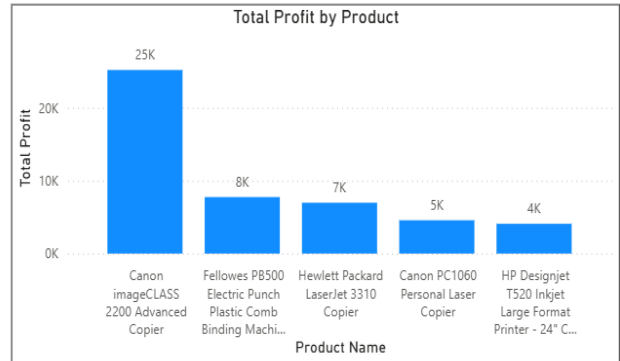
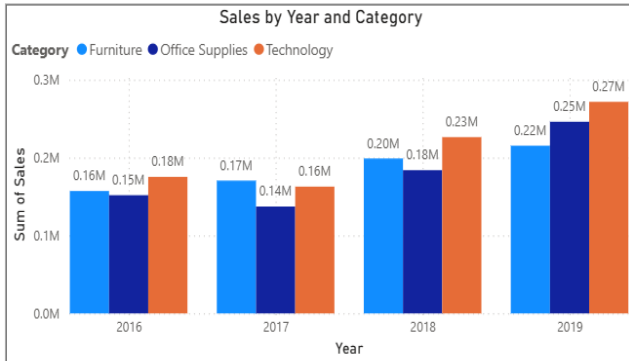
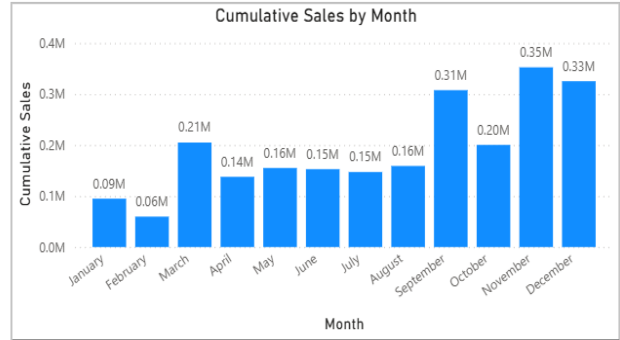
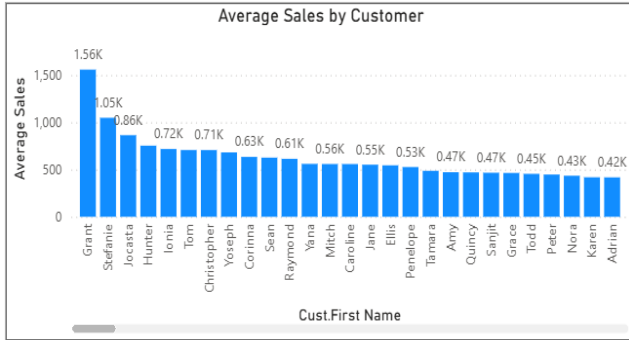


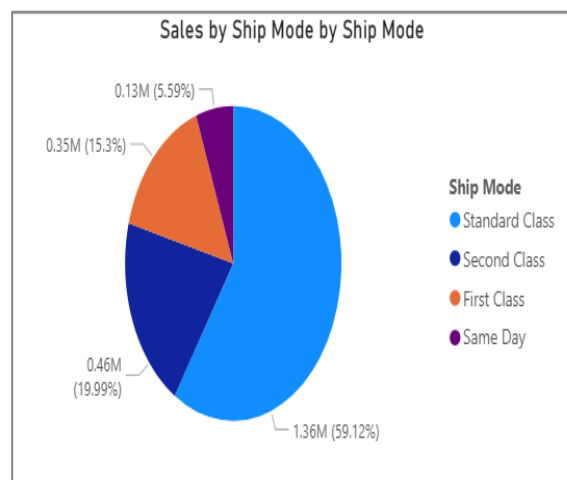
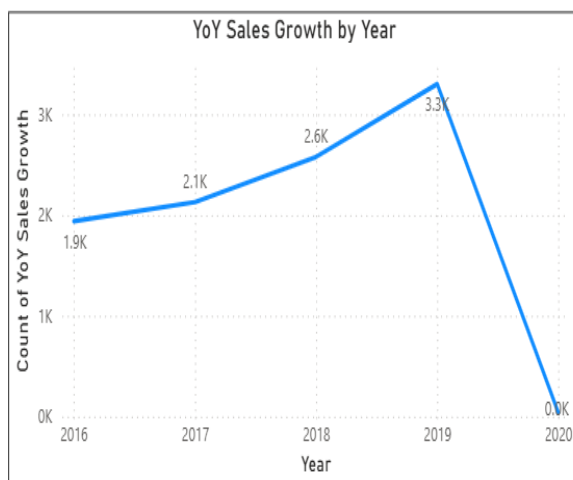
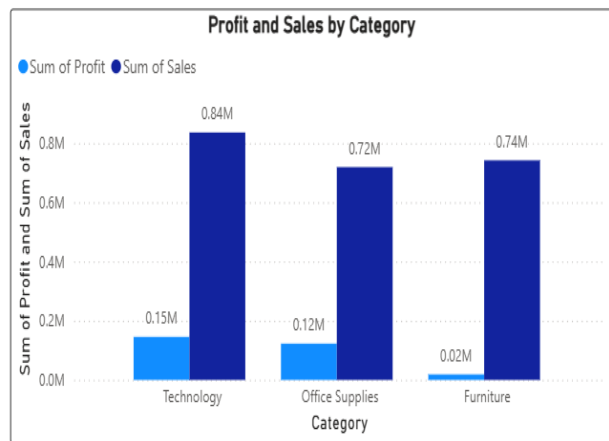
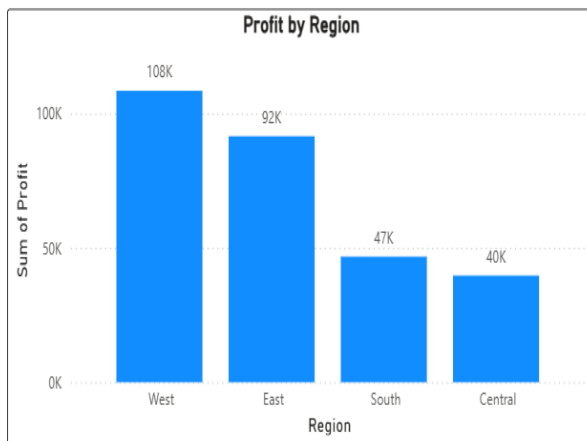
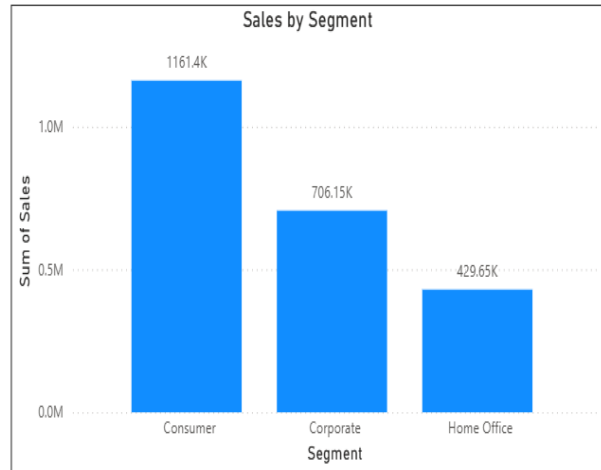
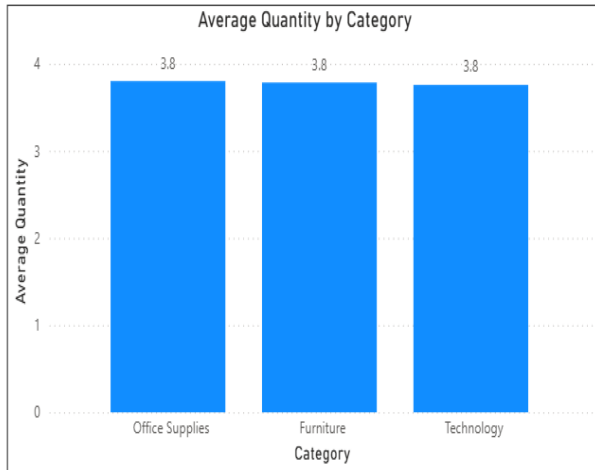
Sales by Region



Total Profit by Month

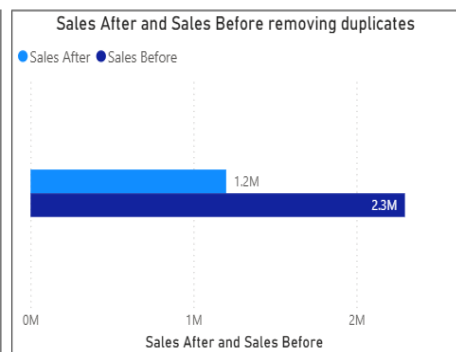
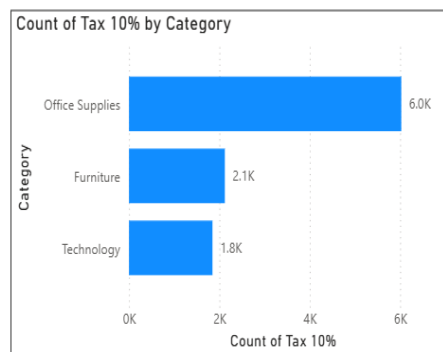
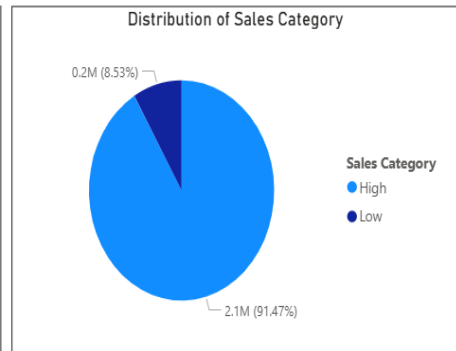
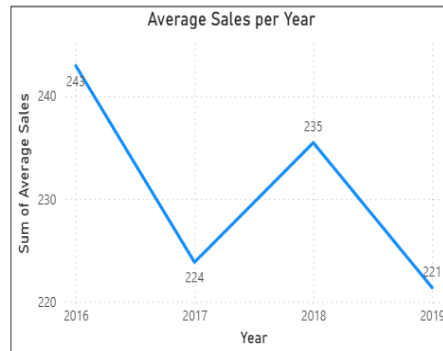






Sales of 2019

7M



Insights from charts and Dashboard:

I. Overall KPI's:

- ✓ Total Sales: 2297.20K
- ✓ Total Profit: 286.40K
- ✓ Total Quantity Sold: 38K
- ✓ Average Discount: 0.09
- ✓ Profit Margin: 0.01K%
- ✓ Sales of 2019: 733.2K

II. Sales Trends:

- ✓ Sales by Year 2016: 0.48M
- ✓ Sales by Year 2017: 0.47M
- ✓ Sales by Year 2018: 0.61M
- ✓ Sales by Year 2019: 0.73M
- ✓ The highest sales is recorded in August Month of the Year 2019 with 1476 sales followed by December Month of the year 2017 with 991 sales.
- ✓ The least sales is recorded in the month of February for the Year 2016 with 13 sales.
- ✓ Standard class shows the highest sales of 1.36M and second class with the least sales of 0.13M.
- ✓ YoY sales growth has shown positive growth trend from 2017 to 2019.
- ✓ The highest sales is observed in the year 2019 in all categories like Furniture, Office Supply and Technology.
- ✓ The West Region gives the highest sales of about 725.46K and the south region gives the lowest sales of 391.72K.
- ✓ The consumer segment contributes to higher sales of 1161.4K and the Home Office contributes to the lowest sales of 429.65K.
- ✓ The sales after removing duplicates shows a significant difference of about 1.1M.

III. Category Insights:

- ✓ Sales by category of Technology, Furniture and Office Supplies are 0.84M, 0.74M and 0.72M respectively, where Technology leads the total sales.

- ✓ Profit by category of Technology, Furniture and Office Supplies are 0.15M, 0.12M and 0.02M respectively, where Technology category yields the highest profit.
- ✓ All three categories have an average quantity of 3.8 units.
- ✓ High value category contributes about 91.47% of total sales.

IV. Sub Category Insights:

- ✓ Among all sub-categories, Storage, Paper, Furnishings, Chairs, Accessories, Labels, Bookcases, Envelopes, Phones and Art are profitable, contributing positively to overall profit.
- ✓ Tables, Binders, and Machines are operating at a **loss**, showing negative profit.
- ✓ Copiers has the highest average profit of 0.82K whereas Tables have the lowest average profit of -0.06K.

V. Regional Insights:

- ✓ West Region has the highest sales with about 725.46K.
- ✓ West Region yields the highest profit with about 108K.

VI. Shipping Mode Insights:

- ✓ Standard class dominates shipment with nearly 60 % share.

VII. Customer Insights:

- ✓ Sean, Tom and John are the top customer with a sale of 41K, 31K and 31K.
- ✓ Top spenders are Grant, Stephanie and Jocasta.
- ✓ William and Xylona shows consistently high monthly sales performance.

VIII. Product Insights:

- ✓ Canon imageCLASS 2200 Copier has the highest sales of \$62K followed by Fellowes PB500 Electric Punch with a sale of \$27K and Cisco TelePresence System with \$23K.
- ✓ Canon imageCLASS 2200 and HP LaserJet 3310 shows the highest profit margins.
- ✓ Copiers have the highest average profit followed by accessories and phone.

IX. Tax Insights:

- ✓ Office Supplies attract the highest number of 10% tax entries.

X. Additional Insights:

- ✓ Sales before removing duplicates and after removing duplicates shows a significant impact of data cleaning.

Conclusion:

This Power BI sales analysis gives a clear picture of the company's performance across several dimensions.

From 2016 to 2019, sales and profit demonstrated steady year-on-year growth, driven primarily by the Technology category and strong contributions from the West and East regions. Standard shipping mode emerged as the most preferred delivery method, accounting for nearly 60% of all sales.

Top-performing customers such as Sean, Tom, and John consistently generated the highest revenue, while products like Canon imageCLASS 2200 and HP LaserJet printers stood out as key profit drivers.

However, certain sub-categories — notably Tables, Binders, and

Machines — recorded negative profits, indicating potential areas for pricing or cost optimization.

Overall, the analysis highlights a healthy upward trend in sales, balanced regional performance, and clear product-level insights.